

Omar Fadlallah

Machine Learning Engineer | AI Engineer
(NLP & RAG Systems)

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Summary

Machine Learning Engineer specialized in NLP and Retrieval-Augmented Generation (RAG) systems, with strong experience building production-grade ML pipelines end-to-end. Proven ability to design, fine-tune, evaluate, deploy, and monitor LLM-based systems with measurable performance and business impact. Hands-on expertise in Transformers, RAG architectures, FastAPI-based serving, MLflow tracking, and drift monitoring.

Education

Bachelor's Degree in Computer Science (Ongoing) at Faculty of Computers and Information, Mansoura University, Mansoura | October 2022 - Present

Grade: 3.7/4.0

Cloud & Deployment (AWS Practical Training – DataCamp): Gained hands-on experience in deploying and managing production-ready infrastructure using core services: EC2, S3, RDS, Lambda, and CloudWatch.

Projects

DeepRAG – MLOps-Ready Retrieval-Augmented Generation System

GitHub: <https://github.com/mirofadlalla/DeepRAG>

- Built an end-to-end production-ready RAG system with evaluation, monitoring, and benchmarking.
- Designed a modular retrieval pipeline using FAISS (IVF), BM25 filtering, and cross-encoder re-ranking.
- Implemented retrieval and generation evaluation (MRR +60%, Recall@K, Precision@K +80%).
- Implemented retrieval evaluation metrics including MRR (+60%), Recall@K, and Precision@K (+80%).
- Added hallucination detection, answer relevance scoring, and grounding validation.
- Integrated MLflow for experiment tracking, metrics logging, and drift monitoring.

Qwen LLM Fine-Tuning — LoRA / QLoRA (Production-Oriented)

GitHub: <https://github.com/mirofadlalla/Qwen-LLM-Fine-Tuning-LoRA-QLoRA>

- Fine-tuned Qwen2.5-1.5B-Instruct using LoRA and QLoRA for efficient adaptation.
- Applied 4-bit quantization and configured LoRA adapters targeting attention layers.
- Trained and evaluated models using Hugging Face Transformers and PEFT.
- Deployed the fine-tuned model with vLLM using runtime LoRA injection for scalable inference.

LLMOps Platform – Production-Grade LLM Prompt Management

GitHub: <https://github.com/mirofadlalla/LLM-Platform-LLMOps-System.git>

- Designed and deployed a FastAPI-based LLMOps platform for prompt versioning and evaluation.
- Built scalable async experiment pipelines using Celery, PostgreSQL, and SQLAlchemy with Alembic migrations.
- Developed a modular evaluation engine with hallucination detection and grounding checks.
- Implemented prompt regression testing and performance tracking in production.
- Containerized the system with Docker and applied production backend best practices.

Internships

Data Science Intern — Talabat, Cairo | On-site | Oct 2025 – Nov 2025

- Selected for Talabat's Data Science Internship program focused on building analytical solutions for business decision-making
- Worked on large-scale datasets to support experimentation, product optimization, and data-driven strategy

- Collaborated with data scientists to design insights for internal stakeholders using SQL and ML.

Microsoft — Software Engineering Intern, Cairo | Jul 2025 – Aug 2025

- Contributed to real-world engineering projects with Microsoft technologies.
- Strengthened production-oriented coding and collaborative development skills.
- Participated in technical workshops and mentorship sessions, strengthening collaborative engineering practices.

Experience

AI Product Engineer — Storyteller Feb 2026 – Present | Remote, Scotland

- Building AI-powered automations for B2B SaaS platform serving global sports brands
- Owning product improvements end-to-end: from identifying problems to shipping solutions
- Developing AI tools to optimize user engagement and debug product behavior

Skills

LLMs & NLP: Transformers, RAG Systems, Prompt Engineering, LoRA / QLoRA, NLTK

Machine Learning & DL: PyTorch, TensorFlow, Scikit-learn, XGBoost

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Power BI

MLOps & Deployment: FastAPI, Flask, Docker, MLflow, Git

Databases: MySQL, MongoDB, FAISS

Programming: Python, SQL, R, C++,

Languages

Arabic – Native

English - Fluent